An alternative to Presuppositional Exhaustification: Supervaluationist RSA[1]

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Abstract. It has been recently argued that exhaustification (the mechanism responsible for deriving scalar implicatures and other exhaustivity effects) should be viewed as generating trivalent meanings, with the effect that scalar implicatures are, in some sense, presupposed (Bassi et al. 2021; Wehbe and Doron 2024). In this paper, I argue that (a) presuppositional exhaustification fails to derive correct felicity conditions in certain cases, and (b) one can replace presuppositional exhaustification with a probabilistic pragmatics account that includes a bivalent exhaustivity operator. Strong Kleene-like behavior is derived on the basis of general principles of rational communication, and adequate felicity conditions are predicted – I call this account, following Cremers et al. (2023), the Supervaluationist RSA account.

Keywords: exhaustivity, scalar implicatures, presupposition, probabilistic pragmatics.

1. Presuppositional Exhaustification

Bassi et al. (2021) propose the following trivalent lexical entry for the exhaustivity operator, which they notate *pex* ('p' is for 'presuppositional').

(1) $$[\![pex_{ALT}(S)]\!]^w = \begin{cases} 1 & \text{if } S \text{ is true in } w \text{ and} \\ & \text{all excludable members of } ALT \text{ given } S \text{ are false in } w \\ 0 & \text{if } S \text{ is false in } w, \\ \text{undefined} & \text{otherwise.} \end{cases}$$

For instance, *pex(Some of the children danced)* is undefined if all children danced, false if no children danced, true if some but not all children danced (I will generally omit the ALT argument). In trivalent approaches to presupposition, the presuppositions of a sentence are equated with its definedness conditions. So *pex(Some of the children danced)* is viewed as *presupposing* that it is not undefined, i.e. that not all of the children danced. The arguments in favor of *pex* (in Bassi et al. 2021 and subsequent work, in particular Wehbe and Doron 2024) are, broadly speaking, of two types. First, *pex* is argued to correctly predict the 'projection' of scalar implicatures and exhaustivity effects. Second, it is argued to correctly predict certain aspects of the felicity conditions of the relevant sentences.

1.1. Strong Kleene projection and *pex*

This trivalent lexical entry needs to be paired with specific trivalent compositional semantics rules. Bassi et al. (2021) adopt a *Strong Kleene semantics* (SK). One immediate advantage of *pex* is the following. If there is an embeddable exhaustivity operator, it is somewhat of a mystery, in a bivalent setting, why scalar implicatures are typically not embeddable under negation, and more generally in downward-entailing contexts. To account for these restrictions, previous literature has typically resorted to constraints on the syntactic distribution of the exhaustivity operator (see, e.g., Fox and Spector 2018). With *pex*, however, no such constraints are needed,

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because, in SK semantics, *pex* has no effect on truth-conditions when it occurs in a downward-entailing environment. This is easily seen in the case of negation. In SK, a negative sentence $\neg S$ is true if S is false, false if S is true, undefined if S is undefined. Given the definition of *pex*, $pex(S)$ is false if and only if S is false. Therefore, $\neg pex(S)$ is true if and only if S is false, i.e. if and only if $\neg S$ is true, and we see that *pex* is in a sense invisible.[2]

1.2. Felicity conditions and *pex*

Bassi et al. (2021) also argue that SIs and exhaustivity effects are typically not part of the 'at issue' content (building on Russell 2006), which would follow from their presuppositional status. At the same time, Bassi et al. (2021) acknowledge that typical scalar items do not seem to trigger infelicity in contexts where the presupposition predicted by *pex* is not contextually entailed. For instance, the answer *I did some of my homework* to the question *What did you do today?* is felt to imply but not to presuppose that I didn't do all of my homework, and the predicted presupposition does not pass the *Wait a minute!*-test. Bassi et al. (2021) posit that the presuppositions triggered by *pex* can easily be *accommodated*, so that their presuppositional status is not directly perceived. Nevertheless they argue that known constraints on accommodation can be seen to be operative in the case of *pex*.

Relatedly, Doron & Wehbe (2024) argue that the presuppositional status of scalar implicatures (and related exhaustivity inferences) can be diagnosed by the fact that they observe a specific constraint on accommodation, the Post-Accommodation Informativity constraint (Doron and Wehbe 2022), which is independently motivated in the case of standard presuppositions:

(2) **Post-Accommodation Informativity constraint (PAI):** A sentence S is not felicitous in a context C if after the presupposition of S is accommodated, S is entailed by the resulting context.

Viewing accommodation as a process whereby the context is enriched with the presupposition of the sentence before being updated by the sentence itself, this constraint can be rephrased as follows, within a trivalent approach to presupposition.

(3) **Post-Accommodation Informativity constraint (PAI):** S is not felicitous in context C if $C \cap Presup(S) \subseteq \{w : [\![S]\!]^w = 1\}$, where $Presup(S) = \{w : [\![S]\!]^w \neq \#\}$

This is in turn provably equivalent to the following constraint:

(4) S is not felicitous in context C if there is no world in C where S is false.

Given the definition of *pex*, this entails the following.[3]

(5) **PAI applied to *pex*:** A sentence of the form $pex(S)$, where S is itself a bivalent sentence, is infelicitous if S is entailed by the common ground.

[2] Bassi et al. (2021) also argue that *pex* has an advantage when dealing with sentences where a *some*-DP occurs under the scope of another one, as in (i), a point we do not discuss in this paper.
(i) Some students solved some of the problems.

[3] By PAI, $pex(S)$ is infelicitous in C if there is no world of C where $pex(S)$ is false. But $pex(S)$ is false just in case S is false. Hence $pex(S)$ is infelicitous in C if there is no world of C where S is false. If S is bivalent, this means that $pex(S)$ is infelicitous in C if every world of C makes S true, i.e. if S is entailed by C

It is then predicted that simple sentences containing a scalar item are infelicitous if their 'literal' meaning is entailed by the common ground. Wehbe and Doron (2024) argue that this prediction is correct, as illustrated by the following kind of example:

(6) To get admission to this group of Hitchcock movie fan, one has to prove that they have watched at least one Hitchcock movie. Sue, Peter and Gloria all belong to the club, and know this. Sue talks with Gloria, and then tells Peter:
  a. #You know what, Gloria saw some Hitchcock Movies.
  b. You know what, Gloria saw some but not all Hitchcock movies.

The infelicity of (6a) follows from the fact that its weak, 'literal' meaning is entailed by the context. The sentence is not saved from infelicity by the fact that the strengthened meaning of this sentence is informative and relevant (as shown by the felicity of (6b)). Specifically, under the parse *pex*(*Gloria saw some Hitchcock movies*), (6a) 'presupposes' *Gloria didn't see all Hitchcock movies*. After accommodation of this presupposition, the context now entails (6a), which is therefore uninformative.

This proposal also has an interesting consequence regarding the interpretation of bare plurals under negation. As is well known (Sauerland et al. 2005; Spector 2007; Zweig 2009), bare plurals under negation receive an inclusive, weak 'one or more' meaning, but tend to trigger a multiplicity implicature when occurring in an upward-entailing context. That is, while (7b) means that Mary didn't read even one book, (7a) suggests that Mary read several books. Spector (2007) and Zweig (2009) proposed that the core meaning of bare plurals is inclusive, and the multiplicity inference is a scalar implicature that results from competition with the singular alternative.

(7) a. Mary read books.
  b. Mary didn't read books.

Spector (2007) noted, however, that bare plurals under negation trigger a kind of modal presupposition – namely, the context must be compatible with the possibility that the multiplicity implicature is true. This is illustrated by the oddness of the following sentence in a typical context where it is not plausible for someone to wear more than one hat at a given time.

(8) #Right now, John is not wearing hats.

Ahn et al. (2020) suggest that the *pex* approach sheds light on this observation. Let us assume that (8) receives the following parse, where *pex(John is wearing hats)* is true if John is wearing several hats, false if he is wearing no hat, undefined otherwise.

(9) NEG(*pex*(John is wearing hats))

(9) is true if John is wearing no hat, false if John is wearing several hats, undefined otherwise. Given the PAI, (9) is felicitous only if it could in principle be false, i.e if it is possible for John to be wearing several hats. Under the assumption that the parse in (9) is the default parse (essentially the view that *pex* is inserted at every scope site), the oddness of (8) is explained.

This example instantiates what we may call 'Magri effects' (Magri 2011): a sentence *S* embedded in a DE-environment is infelicitous if the strengthened reading of *S* (in this case, *John is wearing several hats*) is a contextual contradiction. A more standard example is provided in

(10) (adapted from Magri 2011):

(10) Context: the teacher gives the same grade to every student.
a. #The teacher gave an A to some of her students.
b. #Every teacher who gave a C to some of her students was fired.

(10a) is infelicitous because under the parse $pex(\ldots some \ldots)$ it is a contextual contradiction. (10b) is infelicitous because under the Strong Kleene account, under the parse *[Every teacher who EXH(gave a C to some of her students), ...]*, it 'presupposes' the disjunction of its truth-condition and its falsity condition, i.e: either (a) every student who gave a C to some or all of her students was fired (truth condition), or (b) some teacher who gave a C to some but not all of her students was not fired (falsity condition).[4]

Since the context ensures that no teacher gave a C to some but not all of her students, once the presupposition is accommodated, we know that we are in the (a)-case, i.e. that every teacher who gave a C to some of her students was fired, and the sentence becomes uninformative, and therefore violates the PAI constraint.

So far, *pex*, together with SK and the PAI constraint, seems to do very well. I now turn to a type of felicity contrast not predicted by this combination of assumptions.

2. A felicity contrast not explained by PAI

Consider now the following example, in the specified context, which is a variation on an example discussed in Degen et al. (2015):

(11) Context: Gloria is a scientist. She's investigating a new substance, whose density she does not know. She will perform an experiment by throwing ten marbles, believed to have the same density, into the substance. **She of course expects the marbles to behave uniformly**, i.e. that they will all float or all sink. She's also interested in how the marbles might collide with each other. She will report to her friend Alfred, who knows all of this but does not witness the experiment. **To her surprise, the marbles did not behave uniformly; some sank and others floated.** She hasn't reported this to Alfred yet.
Then Alfred asks her: 'So what happened?' To which Gloria answers: ...
a. #(Something surprising happened.) Some of the marbles sank!
b. (Something surprising happened.) Some but not all of the marbles sank.
c. (Something surprising happened.) Only some of the marbles sank.

Under *pex*, the deviance of (11a) is not explained by PAI. Under the parse $pex(some)$, (11a) 'presupposes' *not all*. Once this presupposition is accommodated, the sentence becomes extremely informative, since it is describing now a very surprising state of affairs, where some but not all of the marbles sank. While a similar example is mentioned in Bassi et al. (2021), it's not clear how one can explain the contrast between (11a) and (11c), if *only* ϕ*'* presupposes

[4]In the SK approach to quantifiers, a sentence of the form *EVERY A B* is true if everything that is an A or is undefined for A happens to be a B, and false if something is defined for A and is an A and is not B. So under the relevant parse, (10b) is true if every teacher who either gave a C to some but not all of her students or gave a C to all of them was fired, and false if some teacher who gave a C to some but not all of her students was fired. It is otherwise undefined

ϕ. Assume that the Common Ground makes it extremely likely that the marbles will behave uniformly. For both (11a) and (11c), the 'presupposition' ('not all' in the case of (11a), 'some' in the case of (11c)) is not entailed by the common ground but is compatible with it, while the assertion ('some' vs. 'not all') nearly contradicts the common ground after accommodation of the presupposition, hence is as informative as it could get.

We can build a comparable contrast between bare plurals and DPs headed by the determiner *several*, as in (12):

(12) What can you tell me about Gloria's educational attainment?
a. #Gloria has college degrees.
b. Gloria has several college degrees.

Under the analysis *pex*(*she has college degrees*), this sentence presupposes that Gloria does not have exactly one college degree (it is false if she has 0, true if she has at least 2). After this is accommodated, one expects her to have no college degree rather than two or more college degrees, since it is very unlikely to begin with to have two college degrees. When adding the assertion, we learn that she in fact has several college degrees. This is highly informative, hence satisfy the PAI in (3). Therefore (12a) should not be deviant, while it in fact is. In order to explain this contrast, it will not do to say that the presupposition is so implausible that it can't be accommodated, because this is simply not true: the presupposition that she has either 0 or 2 college degrees is not implausible, since it is entailed by the proposition that she has no college degree, which is itself plausible if we don't know much about Gloria.

One might posit that whenever we are in such a situation, where the overall meaning is extremely surprising and this is due in part to the presupposition, accommodation is not licensed. But this is incompatible not only with the felicity of (11c), but also with the felicity of (13) in the specified context (Jad Wehbe, p.c.).

(13) [Context: There is an all-boys' school in Bill's town. He doesn't know if Sue has siblings, but he wonders if anyone in her family goes to that school. Mary tells him:] Today I learned something surprising. Sue's sister goes to the all-boys' school.

Here, the second sentence presupposes that Sue has a sister. After this presupposition is accommodated, the second sentence nearly contradicts the common ground. Yet the sentence is perfectly acceptable.

What we seem to observe is the following: when the pragmatically strengthened meaning of a sentence (essentially corresponding to the result of standard *exh* or to the truth-conditions delivered by *pex*) expresses a very surprising proposition but the plain, non-strengthened sentence does not, the sentence feels infelicitous. PAI does not derive this, and these effects do not seem to be expected if SIs and exhaustivity effects behave like presuppositions.[5]

[5]In response to this argument, Doron et al. (2024) have proposed an explanation within the *pex*-approach for the contrast between (11a) and (11c), based on the idea that in the case of (11c), the assertive part of the sentence does not entail its presupposition, whereas it would in the case of (11a). I do not discuss their counterproposal in this paper, which lays out the argument to which they are, in part, responding.

3. A pragmatic account?

Some of the basic contrasts that motivate *pex* as well as the ones we have just discussed might receive a quite natural explanation in terms of a simple probabilistic model of pragmatics, namely a version of the baseline Rational Speech Act model that includes message costs (see, e.g., Goodman and Frank 2016). Critically, though, we will see that this baseline RSA model is not able to account for cases of infelicity where the relevant trigger (a bare plural or *some*) occurs in DE-environment, as in (9) or (10b), and that it faces at least one other problem. But for the time being let us see how far such a probabilistic pragmatic model can go.

In the baseline RSA model, just like in standard neo-Gricean accounts of quantity implicatures, the pragmatic speaker, call her S_1, chooses her message based on the assumption that she is talking to a listener who interprets the message literally. She is bound to say only messages that express propositions she believes, and among such messages, she selects those that achieve a good tradeoff between how informative they are and how costly they are. Crucially, informativity is measured probabilistically: a proposition's informativeness is measured by how *surprising* it is. The lower its prior probability (i.e. the probability the listener assigns to it before receiving the message), the more informative it is.

Due to the tradeoff between informativity and cost, it is possible in principle to dispense with the notion of alternatives. One can envision that every message is a potential alternative, but some are more costly than others. In fact, one can think of the cost of a message as reflecting the probability that the message will come to mind to the speaker.

In this version, just like in the standard neo-Gricean approach to implicatures, the pragmatic speaker (S_1) chooses her message on the basis of its literal meaning, as if she were talking to a non-pragmatic listener L_0, and the pragmatic listener, call her L_1, draws inference about the speaker based on the assumption that the speaker is S_1.[6]

Crucially, we assume that one possible message is the 'null' message, corresponding to saying nothing, whose cost is lower than that of all other messages, and whose meaning is tautological. This is enough to account for the basic fact that PAI derives: if the sentence's literal meaning is contextually uninformative, it is infelicitous. Indeed, in such a situation, the gain in information provided by the sentence for L_0 is null, and not greater than that of the null message, which is less costly. A rational speaker will then always prefer the null message to the relevant sentence, which, therefore, will not be used. The key idea here is that a sentence is predicted to be infelicitous in a given context if it has a very low probability of being used even if it is true.[7]

More generally, a notion of informativity based on surprisal, and not just entailment, can explain how speakers choose between a *some* and a *not all* statements, as noted in Enguehard and Spector (2021); Cremers et al. (2023). Thus consider again the contrast in (6), repeated below:

(14) To get admission to this group of Hitchcock movie fan, one has to prove that they have watched at least one Hitchcock movie. Sue, Peter and Gloria all belong to the club, and know this. Sue talks with Gloria, and then tells Peter:

[6] I will not consider further iterations, i.e. won't go beyond S_1 and L_1.

[7] We must of course allow for a sentence that expresses something very implausible to be potentially felicitous, despite the fact that it has a very low probability of being used since it is probably false. If Trump had picked Hillary Clinton as his secretary of state, it would of course be felicitous and cooperative to report that fact.

a. #You know what, Gloria saw some Hitchcock movies.
b. You know what, Gloria did not see all Hitchcock movies.

In the specified context, the literal meaning of (14a) is completely uninformative. In contrast, (14b) is informative in this context. We find the same contrasts even when the *some* sentence is not totally uninformative, but simply expresses (on its literal meaning) a proposition that is known to be very likely true, while the *not all* sentence expresses a much more surprising proposition. This is illustrated in (15) (based on Enguehard & Spector 2023):

(15) At an international conference, one expects talks to be in English. Gloria reports:
a. #Some talks were in English.
b. #Many talks were in English.
c. Not all talks were in English.

In the baseline RSA framework, there is a tradeoff between informativity and cost. In general, in a *some but not all* situation, if the prior probability distribution does not make *some* very underinformative, a speaker might prefer *some* over *not all* and *some but not all* because *some* is the least costly message, *not all* is no more informative than *some* and is more costly, and the gain in informativity brought by *some but not all* is not worth its cost. That is, the classical 'symmetry problem' is solved by appealing to cost. Even if *some but not all* is an alternative, it is more costly than *some*, unlike *all*, so the listener reasons in the following way: the speaker who said *some* is more likely to be in a *sbna*-situation than an *all*-situation, since in the *all* situation, *all* would have been a better message to use, given that it is more informative and no more costly, while in the *sbna*-situation, *some but not all* is not necessarily a better message than *some*, due to its extra cost. However, while cost can 'break' symmetry, so can informativity, measured probabilistically. That is, in a biased context where *not all* and *some but not all* are much more informative (i.e. surprising) than *some*, speakers will prefer *not all*, or even *some but not all* to *some* if they believe *some but not all*, making it very unexpected for a speaker to say *some*, as in (15).

Competition between *some* and *some but not all* can also explain our new data point in (11), repeated below.

(16) Same context as in (11): it is expected that the marbles either all sank or all floated. In fact some but not all of them sank.
a. Alfred: So what happened?
b. #Some of the marbles sank.
c. Some but not all of the marbles sank.

A purely pragmatic story based on probabilistic informativity and cost can account for this contrast. In a nutshell, in this context, *some* is somewhat informative, but also incredibly misleading, since if Alfred updates his beliefs with *some*, he will assign a very high probability to *all*, which is the opposite of what the speaker wants to communicate. Furthermore, *some but not all* is now incredibly surprising (hence informative), since the prior probability that is is true is close to 0. Therefore a rational speaker who chooses her message based on a tradeoff between informativity and cost will in fact prefer *some but not all*, whose extremely high informativity dwarfs its cost disadvantage, especially given how misleading *some* is. Formally, one can show, in the baseline RSA model that includes as messages *some*, *all*, *some but not all*, with a higher cost for the latter message, that the probability of using *some* to express *sbna* decreases as the the prior probability that *sbna* is true decreases, and that a fully rational speaker will simply never use *some* if the probability that *sbna* is true is small enough.[8]

4. Limitations of the pragmatic account

The pragmatic approach just sketched faces several limitations. To begin with, it is unable to predict 'Magri effects' in DE-environments, illustrated again with the following example, repeated from (10):

(17) Context: the teacher gives the same grade to every student.
Every teacher who gave a C to some of her students was fired.

In the specified context, this sentence is contextually equivalent to 'Every teacher who gave a C to all of her students was fired'. This is both informative and in no way misleading if true. The pragmatic account thus does not account for the deviance of (17).

Similarly, the pragmatic approach does not provide any clear explanation for why the following sentence is infelicitous in a context where Gloria is not expected to have several college degrees:

(18) #Gloria does not have college degrees.

Furthermore, as discussed in detail in Cremers et al. (2023), building on Degen et al. (2015), a purely pragmatic approach based on a tradeoff between probabilistic informativity and cost predicts unattested *anti-exhaustivity effects*. That is, in the following dialogue, given the context, it predicts the answer in (19b) to be felicitous and, in fact, an excellent way to convey that the two marbles sank, contrary to facts.

(19) Context: Gloria is experimenting with just two marbles, a blue one and a red one. She throws them both into an unknown substance, and it is expected that they will either both sank or both float. She then reports the results of this experiment to her friend Alfred:
a. Alfred: What happened?
b. #The blue marble sank.

The reason for this prediction is that the literal meaning of (19b), given the context, makes it highly probable that both marbles sank, and it is furthermore less costly than *The blue and the red marbles sank*. Furthermore, if, in fact, contrary to expectations, only the blue marble had sunk, the speaker would have had a very strong incentive to say it explicitly, since (19b) is so poor at doing that. (19b) therefore suggests that it is not the case that in fact only the blue marble sank.[9] The pragmatic account discussed above therefore makes two wrong predictions about (19b): it predicts it to be felicitous, and it predicts it to convey, in the relevant context, that both marbles sank.

[8] See Enguehard and Spector (2021); Cremers et al. (2023) for a formal analysis of the predictions of the baseline RSA model.
[9] See Cremers et al. (2023) for a mathematical derivation of this bad prediction in the baseline RSA model.

5. A supervaluationist RSA account: the intuition

I now present a system which, unlike the pragmatic account discussed above, includes a grammatical exhaustification mechanism. This system inherits the successful predictions of both the pragmatic account and *pex*, including those that lie beyond the reach of either approach alone.The proposal is based on a modification of the supervaluationist RSA account proposed in Spector (2017) and Cremers et al. (2023). I present here in an informal way. The mathematical model is presented in section 6.

The core idea is the following. I assume a 'standard', bivalent exhaustivity operator *exh*. Every sentence is underspecified between a parse with *exh* and one without, which creates a coordination problem between the speaker and the listener. The way this underspecification is pragmatically resolved gives rise to a Strong Kleene projection pattern, and furthermore explains all the felicity contrasts we have discussed. Specifically, a *safe* speaker will use an underspecified sentence only if, *under all possible specifications*, the resulting fully specified parse satisfies Gricean maxims. The motivation for such a policy is a form of risk aversion: it is not safe to use a sentence under some intended parse if under some other parse that the listener might reconstruct, things would go terribly wrong.

Consider now a simple sentence like *Some of the guests left* and its negation *It's not the case that some of the guests left*. Both are underspecified between a parse with EXH (below negation in the case of the negative sentence) and without:[10]

(20) a. (EXH) Some of the guests left.
  b. NEG (EXH) Some of the guests left.

Our proposal is that for each of these two sentences

1. Both parses should express **true** propositions ('Truth on all parses')
2. Both parses should be informative ('Informativity')
3. Both parses should represent a reasonable tradeoff between informativity and cost, and not be misleading ('Tradeoff')

The first constraint ('Truth on all parses') gives rise to a Strong Kleene-type projection. Let me make this point formally precise in the case of propositional logic. Let us assume a propositional language where sentences are evaluated with respect to trivalent valuations (functions from atomic sentences to $\{0,\#,1\}$). For any atomic sentence p, we associate two special atomic sentences notated p^+ and p^- and impose the following meaning postulates on these two propositions: for any valuation v, p^+ is true in v if p is true in v, p^+ is false otherwise (i.e., if p is undefined or false in v); p^- is true in v if p is true or undefined in v, false otherwise (i.e., if p is

[10] I ignore here the parse *EXH(NEG (EXH (some of the guests left)* which Fox & Spector (2018) argued expresses the proposition that all of the guests left. Importantly, we argued that this parse requires narrow focus on *some*. More generally, I do not discuss here how the proposal made in this paper interacts with the previous literature on recursive exhaustification (i.e parses in which *EXH* occurs under the scope of an other occurrence of *EXH*). In this particular case, I assume that the higher *EXH* in the parse *EXH(NEG(EXH ...))* is vacuous, in the absence of special prosodic marking. This could be derived in a theory where EXH blocks the projection of alternatives from its scope and cannot itself be deleted when computing alternatives, in the absence of special prosodic marking. As discussed in Bade and Sachs (2019), such assumptions are not compatible with a number of works in which recursive exhaustification is used, a complication that we leave for further research.

false in v). By construction, p^+ and p^- are bivalent sentences (they are never undefined). Let S be a propositional logic sentence. Let $\{p_1,\ldots,p_n\}$ be the atomic sentences that occur in S (possibly multiple times). We consider all sentences that result from replacing each occurrence of any p_i with either p_i^+ or p_i^- (allowing for non-uniform substitution in case of multiple occurrences; one occurrence of p_i might be replaced with p_i^+ while another one is replaced with p_i^-). Call the set of all such sentences the *bivalent variants of S*. Then S is true in v according the SK semantics if and only if every bivalent variant of S is true in v, S is false according to the SK semantics if and only if every bivalent variant of S is false in v, and S is undefined according to the SK semantics if and only if the bivalent variants of S do not all have the same truth-value in v. Now, consider a sentence S that contains a unique occurrence of *pex(some)*, call it $S(pex(some))$. The bivalent variants of $S(pex(some))$ are obtained by replacing $pex(some)$ with *some* or *EXH(some)*, which shows that 'truth on all parses' corresponds to 'truth according to the SK semantics'. This point generalizes when there are multiple occurrences of *pex*. It can also be extended to a language with quantifiers (using a similar construction), though I do not show this here. In other words, 'truth on all parses' gives rise to the same projection patterns as those expected under the *pex*-view.[11]

In particular, scalar items are predicted to be strengthened in upward-entailing environment and not in downward-entailing environments:

(21) Mary solved some of the problems.
  a. Parse without *exh*: $\exists$
  b. Parse with *exh*: $\exists\neg\forall$
  c. Both parses must be true: $\exists\neg\forall$

(22) It is not the case that Mary solved some of the problems.
  a. Parse without exh: $\neg\exists$
  b. Parse with embedded exh: $\neg exh(some) = \neg\exists \vee \forall$
  c. Both parses must be true: $\neg\exists$

(23) Gloria has pets.
  a. Parse without *exh*: Gloria has at least one pet
  b. Parse with *exh*: Gloria has two pets or more
  c. Both parses must be true: Gloria has at least two pets

(24) Gloria does not have pets.
  a. Without *exh*: Gloria has no pets
  b. With embedded *exh*: Gloria does not have two pets or more
  c. Both parses must be true: Gloria has no pets.

The second constraint (Informativity) captures the effects predicted by the PAI constraint together with *pex*-accounts, for all the cases we have discussed. In particular, in all the cases where the relevant parses stand in an entailment relationship, the constraint amounts to the requirement that the logically *weaker* parse be informative. When *some* occurs in an upward entailing context, this boils down to the requirement that the literal meaning of the sentence

[11] A caveat: in the *pex* view, it is as if we only consider the parses where EXH occurs, if at all, where *pex* occurs, i.e. the *pex* view allows for cases where a scalar item is simply ignored, if it is not under the scope of *pex*. However, most *pex*-based accounts (including Bassi et al. 2021) posit that *pex* is obligatory at every scope site. This point is potentially relevant to the discussion of *some*-under-*some* sentences, cf. footnote 2.

be informative, i.e. not already entailed by the context. So for instance, *Gloria saw some Hitchcock movies* is predicted to be infelicitous if the context already entails that Gloria saw some or all of Hitchcock movies. But when *some* occurs in a downward-entailing environment, the requirement becomes that the reading with embedded implicature (the weaker reading) be informative. Consider for instance (10b), repeated below in (25):

(25) Every teacher who gave a C to some of her students was fired.

In a context where it is already known that every teacher gave the same grade to all of her students, the parse with an embedded EXH is entailed by the context. Indeed, this parse expresses the proposition that every teacher who gave a C to some but not all of her students was fired, but in such a context, it is known that there are no such teacher, making the sentence vacuously true. The informativity constraint therefore correctly predicts that (25) is infelicitous in such a context. Generally speaking, it is able to capture Magri effects. Consider in this light (8), repeated as (26):

(26) #John is not wearing hats.

The parse with an embedded EXH expresses the proposition that John is not wearing several hats. But this is uninformative in a standard context in which it is expected that people wear at most one hat. Hence the sentence is predicted to be infelicitous, as desired.

Finally, the third principle ('Tradeoff') accounts for the new data discussed in section 2. Because the principle requires that both parses satisfy a good tradeoff between informativity and cost, the account provided in section 3 for the marbles example in (16) carries over. Consider also (12) again, repeated below:

(27) What can you tell me about Gloria's educational attainment?
  a. # Gloria has college degrees.
  b. Gloria has several college degrees.

(27a) is deviant and, in particular, cannot be used felicitously in the context of this question to convey that Gloria has several college degrees. The tradeoff constraint explains this fact in the following manner. On its literal reading (parse without EXH), (27a) means that Gloria has at least one college degree. Now, (27b) is much more informative than this proposition, not just because it entails it, but because having several college degrees is highly surprising. Hence despite its higher cost, if Gloria in fact has several college degrees, (27b) is much better than (27a) under its literal reading. This is all the more so that the proposition expressed by the literal meaning of (27a) is also misleading since, given the background knowledge that it is very rare to have several college degrees, it will lead the listener to be believe that Gloria probably has exactly one college degree. This is therefore a case where a cooperative speaker should use (27b), despite its higher cost, which explains the infelicity of (27a) (which is also not usable if Gloria in fact has exactly one college degree, due to the 'Truth on all Parses' constraint).

6. A probabilistic supervaluationist model

I now present a simple RSA model which provides a precise, explicit implementation of the ideas discussed in the previous section. This model is a modified version of the supervaluationist RSA account presented in Spector (2017) and Cremers et al. (2023), and I only consider the first level pragmatic speaker (S) and the first level pragmatic listener (L). The listener is characterized by a probability distribution P over worlds, which she updates after receiving a message. We first define the utility of a message σ for the first-level pragmatic speaker (the only speaker we will consider here) relative to a specific meaning ϕ for this message:

$$U(\sigma,\phi,w) = \log(P(w|\phi)) - cost(\sigma)$$

This captures the following idea: if the speaker wants to communicate a world w, the utility of a message σ understood as meaning ϕ is (a) an increasing function of the probability that the listener assigns to w after she updates her distribution with ϕ, and (b) a decreasing function of the cost of the message. Importantly, if ϕ is false in w, $P(w|\phi) = 0$, hence $\log(P(w|\phi)) = -\infty$, and therefore the overall utility is itself $-\infty$. This makes it impossible to use a false message, which captures Grice's maxim of Quality.

Now, what should the speaker do when there is uncertainty as to how the message is going to be understood, due to the existence of multiple parses? There are several conceivable ways to deal with such uncertainty. If the speaker is extremely risk-averse, which is what we assume here, the speaker will consider the outcome of her choice *in the worst case*, i.e. the case where the listener interprets the message under the parse that provides the least utility. This leads to the so *MaxiMin rule*, whereby the speaker's policy is to maximize utility under the worst possible interpretation of each message: for each message, the speaker acts as if its utility is the minimum of all the utilities that this message could have under its different parses. We thus define the utility of a message relative to a world as being the *minimum* of all the utilities provided by this message in this world across all its possible parses.[12]

(28) Suppose σ has parses expressing propositions $\phi_1, \phi_2, \ldots, \phi_n$. Then:
$U(\sigma,w) = min(U(\sigma,\phi_1,w), U(\sigma,\phi_2,w), \ldots, U(\sigma,\phi_n,w))$

As usual in RSA models, the pragmatic speaker (S) is modeled as being only approximately rational. Instead of picking a message that maximizes utility with probability 1, the speaker is more likely to pick optimal messages than non-optimal ones, but may pick non-optimal messages with a non-null probability. A parameter – a positive real number notated λ – determines how rational the speaker is. The higher λ is, the more likely the speaker is to pick an optimal message. Specifically, the pragmatic speaker's probability of choosing a message σ in a given world w is given by the following equation (where σ' ranges over all possible messages):

$$S(\sigma|w) = \frac{\exp(\lambda U(\sigma,w))}{\sum\limits_{\sigma' \in \mathscr{M}} \exp(\lambda U(\sigma',w))}, \text{ where } \mathscr{M} \text{ is the set of all possible messages.}$$

Note that when σ is false under one of its parses in w, $U(\sigma',w) = -\infty$, and so $\exp(\lambda U_1(\sigma,w)) = 0$, resulting in $S(\sigma|w) = 0$.[13] This implements the principle of *Truth on all parses*: the speaker never uses a message if it is false under one of its parses.

[12] This is where the present proposal departs from the one in Spector (2017) and Cremers et al. (2023), where the utility of a message was defined as its expected utility across all its possible parses, while here it is defined as the minimum of all the utilities achieved by a message across all its parses. The model in Spector (2017) and Cremers et al. (2023) also derive the 'Truth on all parses' principle, hence SK-like projection, but make the other predictions less robust than the present version.

[13] For the denominator to be positive, it is necessary and sufficient that for every world, there is at least one message which is true in that world under a certain parse. We can ensure that this is necessarily the case by including a *null message* which is true in every world.

The pragmatic listener L is defined as a Bayesian agent who knows that she receives a message from S and updates her belief based on this knowledge, i.e. based on how likely S was to send a given message in different worlds.[14] In particular, she interprets a sentence as being true on all its parses due to her knowledge that the speaker's probability of using a message is 0 as soon as it is false on one of its parses.

In order to make the model explicit, we need to specify a set of possible worlds, messages, message costs, and parses for each message. Importantly, we will always assume that there is a *null message*, which is always true, and has a null cost (it can be thought as representing silence). Let us consider a case with 3 possible worlds, $w_{\neg\exists}$ ('No marbles sank'), $w_{\exists\neg\forall}$ ('Some but not all of the marbles sank'), $w_{\forall}$ ('All of the marbles sank'), and 6 messages: *sm* ('some'), *nall* ('not all'), *sbna* ('some but not all'), *all*, *null*, with the following assumptions about costs: $c(null) = 0, c(sm) = c(all) < c(nall) < c(sbna)$. Finally, while *null*, and *sbna* have only one parse (making them unambiguous) *sm* can be parsed as meaning either $\{w_{\exists\neg\forall}, w_{\forall}\}$ (standard literal meaning) or as meaning $\{w_{\exists\neg\forall}\}$ (exhaustified meaning), while *nall* can be parsed as meaning either $\{w_{\exists\neg\forall}, w_{\neg\exists}\}$ (literal meaning) or $\{w_{\exists\neg\forall}\}$ (exhaustified meaning).

In this setup, it can be shown that if a sentence S is uninformative under one of its parses ϕ, i.e. $P(\phi$ is true) is very close to 1,[15] then the utility of S is less than that of the null message, so a sufficiently rational speaker will not use it with any significant probability. This essentially captures the principle that all parses should be sufficiently informative, and thereby the effects of the PAI constraint in the *pex*-based approach. It can also be proved that if $P(w_{\exists\neg\forall})$ very small compared to $P(w_{\forall})$,[16] S will not use the message *sm* with any significant probability if she in fact believes that the actual world is $w_{\exists\neg\forall}$, but will instead choose *nall* or *sbna*. In the marbles example described in (11), where the worlds $w_{\neg\exists}$ and $w_{\forall}$ are equiprobable and together have a probability close to 1, the model predicts that a speaker who believes $w_{\exists\neg\forall}$ will not say either *sm* or *nall*, but will choose *sbna*. The reason is that in such a case the utility of *sm* to communicate $w_{\exists\neg\forall}$ is much lower than that of *sbna*, because on its worse parse *sm* is compatible with $w_{\forall}$ and, given the prior, would lead the listener to put nearly all of her credence on $w_{\forall}$ rather than on $w_{\exists\neg\forall}$.

I implemented the model with the following specifications: $\lambda = 10$, $c(null) = 0, c(sm) = c(all) = 0.25, c(nall) = 0.75, c(sbna) = 1.25$. I considered three prior probability distributions: a) a flat distribution ($P(w_{\neg\exists}) = P(w_{\exists\neg\forall}) = P(w_{\forall}) = \frac{1}{3}$), b) a distribution in which the literal meaning of *sm* is close to being completely uninformative ($P(w_{\neg\exists}) = \frac{1}{50}, P(w_{\exists\neg\forall}) = P(w_{\forall}) = \frac{49}{100}$), and c) a distribution in which $w_{\exists\neg\forall}$ has a very low probability while the two other worlds are equiprobable, which corresponds to the marbles example in (11) ($P(w_{\exists\neg\forall}) = \frac{1}{10}, P(w_{\neg\exists}) = P(w_{\forall}) = \frac{9}{20}$). Under the flat distribution, the predictions for the speaker and the listener are as described in the following tables, showing that *sm* is, as expected, used with a high probability (0.79) to convey $w_{\exists\neg\forall}$, and interpreted as conveying this as well (just like *sbna*

[14] $L(w|\sigma) = \dfrac{P(w) \times S(\sigma|w)}{\sum\limits_{w' \in \mathscr{W}} P(w') \times S(\sigma|w')}$, where $\mathscr{W}$ is the set of all possible worlds.

[15] We cannot always set $P(\phi$ is true) to exactly 1, because this would entail that non-ϕ worlds receive a null probability, which can make some messages uninterpretable. For instance *No* is necessarily false if $P(\{w_{\exists}\neg\forall, w_{\forall}\}) = 1$, which makes the message *No* uninterpretable by L_0.

[16] How small exactly depends on precise cost values and on the rationality parameter

and *nall*, which however have a low probability of being used).[17]

Table 1: Speaker's strategy under flat priors

	sm	*all*	*sbna*	*nall*	*no*	*null*
$w_{\neg\exists}$	0	0	0	0	1	0
$w_{\exists\neg\forall}$	0.79	0	0.04	0.01	0	0.17
$w_{\forall}$	0	1	0	0	0	0

Table 2: Listener's interpretation, flat priors

	sm	*all*	*sbna*	*nall*	*no*	*null*
$w_{\neg\exists}$	0	0	0	0	1	0
$w_{\exists\neg\forall}$	1	0	1	1	0	1
$w_{\forall}$	0	1	0	0	0	0

In case b), i.e. when the priors are such that the literal meaning of *sm* is extremely uninformative, we find that *sm* is essentially not used. If the speaker believes that the actual world is $w_{\exists\neg\forall}$, she either stays silent or uses *nall*, which seems intuitively accurate (Table 3).

Table 3: Speaker's strategy when $P(w_{\neg\exists}) \simeq 0$

	sm	*all*	*sbna*	*nall*	*no*	*null*
$w_{\neg\exists}$	0	0	0	0	1	0
$w_{\exists\neg\forall}$	0.06	0	0	0.3	0	0.64
$w_{\forall}$	0	0.99	0	0	0	0.01

Finally, in the marbles scenario (case (c) above), where $w_{\exists\neg\forall}$ has a very low prior probability compared to the two other worlds, *sm* is likewise not used.[18] In order to convey *nall*, the speaker is predicted to use *sbna*. This can be seen in Table 4.

Table 4: Speaker's strategy when $P(w_{\exists\neg\forall}) \simeq 0$

	sm	*all*	*sbna*	*nall*	*no*	*null*
$w_{\neg\exists}$	0	0	0	0	1	0
$w_{\exists\neg\forall}$	0	0	1	0	0	0
$w_{\forall}$	0	1	0	0	0.	0

7. Gradience

A potential benefit of this approach is that it makes the notion of informativity gradient, thereby predicting PAI type patterns in cases where, in *pex*-terms, the proposition being expressed is not totally uninformative after presupposition accommodation, but merely extremely unsurprising. Consider in this light the following data:

(29) Context: in this country, half of the population does not attend college. The other half attends college. About 10% of them drop out and do not obtain any degree. Among the 40% who get a college degree, the huge majority gets, as expected, exactly one degree, but a very tiny minority (1%) happens to graduate with several college degrees.
  a. #Gloria has college degrees.
  b. #Gloria doesn't have college degrees.
  c. Gloria has several college degrees.
  d. Gloria doesn't have a college degree.

As already discussed, the deviance of (29a) is explained in the same way as that of (16b), i.e. as derived from the tradeoff between informativity and cost. Since having several college degrees

[17] Values are rounded to the 2nd decimal, so a value of 1 typicallly means a value very close to 1, but possibly less than 1 by a very small amount.
[18] Values are rounded, so the actual probability is not 0, but is insignificant.

is *a prior* highly unlikely, the parse under which (29a) means that Gloria has at least one college degree is much less informative than (29c) and so loses to it, despite (29c)'s higher cost.

Now, the deviance of (29b) is not directly accounted for under the *pex*-account. In the *pex*-based approach, after accommodation of the 'presupposition' that Gloria does not have exactly one college degree, the context contains both worlds where she has no college degree and worlds where she has several college degrees. So the 'assertive' content of (29b), namely that Gloria has no college degree, is still informative in a strict sense, as it excludes the latter worlds. In our approach, informativity is gradient (since it is measured probabilistically), and the oddness of a sentence like (29b) is itself expected to be gradient: the more unexpected it is for Gloria to have several college degrees, the more deviant will (29b) feel. The logic of the account is the following. One of the parses for (29b) means that Gloria does not have several college degrees. While this proposition is not strictly speaking uninformative, it is however extremely uninteresting in a world where Gloria's having several college degrees is extremely unlikely to be true. For this reason, the worst parse of (29b) has very low utility compared with the two parses that (29d) has: even the worse parse of (29d), namely the one that says that Gloria does not have exactly one college degree, is highly informative, as it rules out a world which has a significant probability. (29b) is therefore expected to lose to (29d). On the *pex*-based approach, one possibility is to revise the PAI constraint must so as to make it a gradient constraint that refers to prior probabilities (as independently noted by Yizhen Jiang and Haoming Li, p.c.):

(30) **Probabilistic PAI**
Let P be the probability distribution representing the listener's beliefs about some topic of interest, known to the speaker. Then the closer $P(S \text{ is true } | S \text{ is defined })$ is to 1, the less felicitous S is.

Another possibility (Danny Fox, p.c.) is that the notion of Common Ground that underlies the *pex*-account (and presupposition theory in general) has to be understood as representing a form of *pretense* on the part of speakers: speakers act *as if* something is common knowledge, and listeners and speakers constantly negotiate what they take for granted, but actual common knowledge is most often beyond reach. Such a view is, arguably, independently motivated (including, among other things, by the way we think about presupposition accommodation). One can then view the oddness of (29b) as reflecting the fact that, in practice, speakers do *as if* having several college degrees were impossible, because it is so atypical. A potential issue with this response is that even when the possibility of having several college degrees is explicitly mentioned in previous discourse, while also presented as extremely rare and atypical, a sentence like (29b) remains quite deviant: if we make the 'context' in (29) part of the preceding discourse, (29b) remains odd.

8. QUD sensitivity

The current proposal is incomplete, as it does not address the role of context in licensing exhaustivity effects, and, specifically, relevance to an underlying Question Under Discussion (QUD). For this reason, the current proposals predicts exhaustivity effects to be obligatory, while there is ample evidence that they are modulated, among others, by the underlying QUD. There are at least two natural ways to incorporate QUD dependence in the model, which are not mutually incompatible. One consists in assuming, with much of the existing literature, that the alternatives with respect to which exhaustivity is computed are constrained to be relevant. EXH is in fact a focus-sensitive operator which takes two arguments: a set of alternatives and its 'prejacent'. The first argument is expected to be constrained by relevance considerations, previous discourse, etc. A second, very natural way to incorporate QUD dependence consists in a refinement of the proposed RSA model along the following line (as is done in the related model proposed in Spector 2017 and Cremers et al. 2023). Instead of modeling the speaker as wanting to communicate the actual world, we model it as wanting to communicate the true answer to the QUD, as well as the identity of the QUD she is answering. Listeners then are assumed to make joint inferences about both the world and the intended QUD. As discussed in Cremers et al. (2023), a consequence of such a view is that scalar implicatures and exhaustivity effects are predicted to be *obligatory if relevant*, and a sentence such as *Some of the students came* then triggers the inference that either not all students came, or it is not relevant whether all students came. Buccola and Haida (2019) precisely argue for such a view. The detailed exploration of such an extension, and its different variants, as well as a detailed discussion of their predictions (including for *some*-under-*some* sentences, cf. fn. 2), is left for future work.[19]

9. Conclusion and open issues

I have argued that the good predictions of the *pex*-approach can be captured in a system where the exhaustivity operator receives its standard bivalent semantics, but uncertainty about parses (i.e. with or without EXH) gives rise to a Strong Kleene-like projection pattern, given a maximally risk-averse speaker. This approach, once embedded in a probabilistic pragmatics model, was also shown to make additional correct predictions which are beyond the reach of the *pex*-account coupled with the PAI constraint.[20] Among the many issues that are left unresolved are the role of QUDs and focus marking (esp. in cases where exhaustivity effects are triggered in the scope of negation cf. Fox & Spector 2018). Another central issue, which is shared with approaches which assume exhaustivity to be obligatory at every scope site (Magri 2011, Bassi et al. 2021, Del Pinal 2021, a.o.) is the following (Moysh Bar-Lev, p.c.). Consider a sentence as simple as *Mary came and John came*, as an answer to *Who came among Mary, John and Sue?*. If EXH is present at every scope site, there is a parse in which the first conjunct is under the scope of EXH, and is thereby interpreted as 'Only Mary came among Mary, John and Sue', making the whole sentence contradictory.[21] Clearly, to make the account complete (but also some other proposals on the market which partly cover the same empirical ground), we need an explicit theory which restricts the available parses.

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[19]Thanks to Yizhen Jiang and Haoming Li for relevant discussions which convinced me that such an extension might face non-trivial challenges.

[20]But see Doron et al. (2024) for an interesting counterproposal.

[21]One could consider the possibility that alternatives are freely pruned in order to avoid contradiction. But if we want to preserve, say, Magri's (2021) account, we need to assume that alternatives that are contextually equivalent to the prejacent cannot be pruned. So in a context where it is known that John came if and only if Mary came, *Mary came and John came* should feel contradictory if EXH is obligatorily present at every scope site.

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